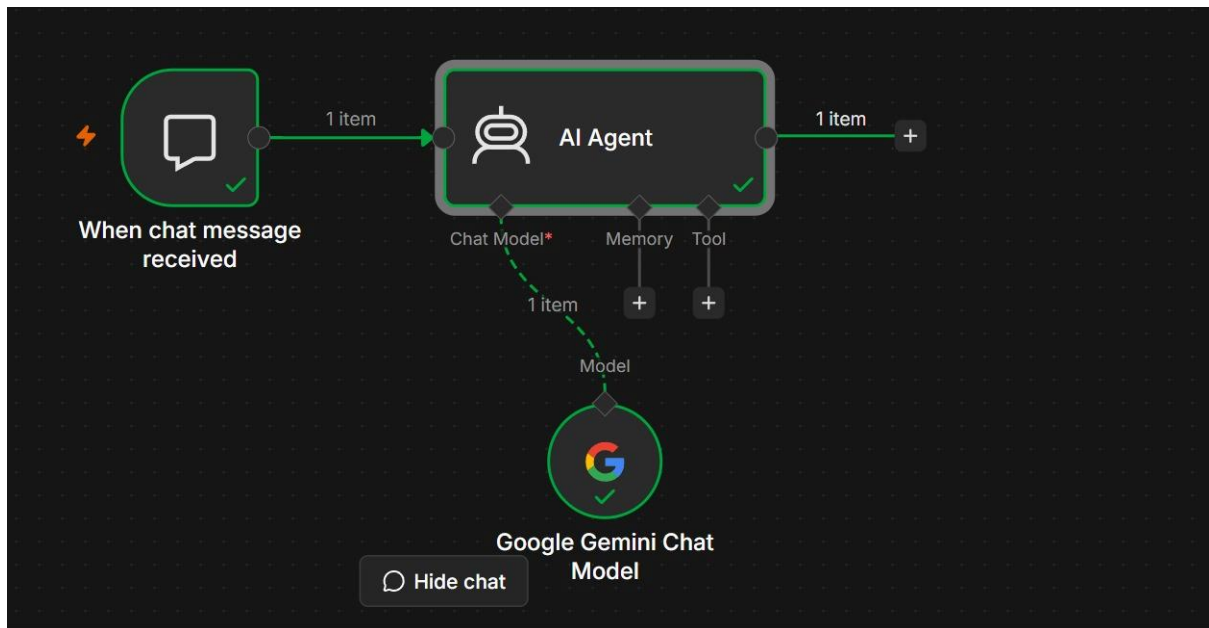


PROJECT-19

AI Hashtag Generator



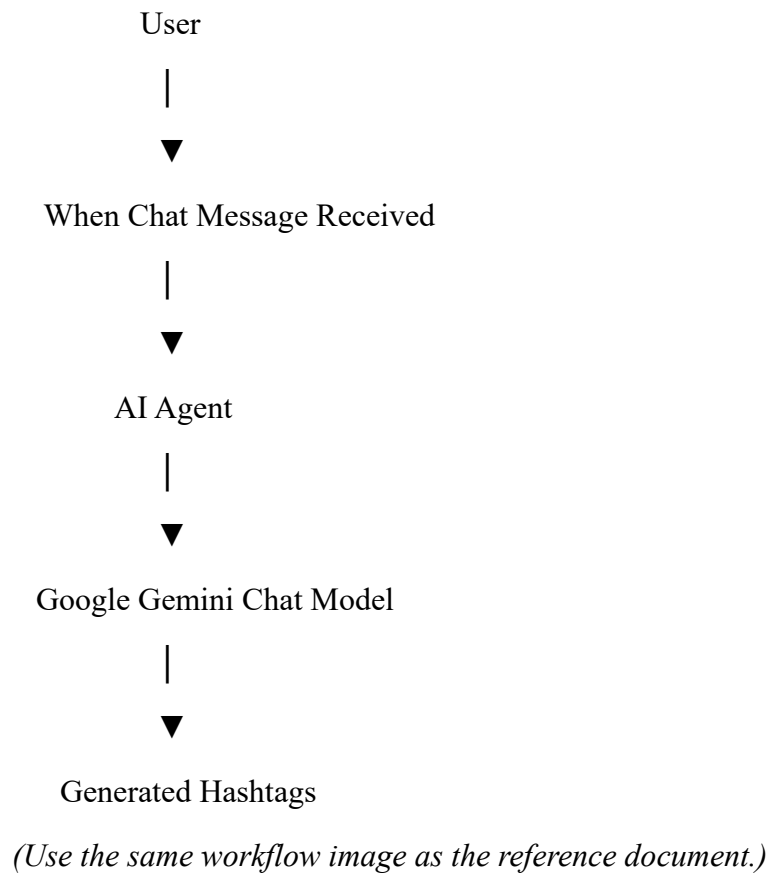
Project Objective

The objective of this project is to build an **AI-powered Hashtag Generator** that creates relevant, trending, and high-performing hashtags based on a user's topic, keyword, or social media post.

The AI Agent can generate:

- Instagram hashtags
- LinkedIn hashtags
- X (Twitter) hashtags
- Facebook hashtags
- YouTube hashtags
- Marketing hashtags
- Business hashtags
- Technology hashtags
- Travel hashtags
- Fitness and lifestyle hashtags

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's request
2	AI Agent	Understands the topic and prepares the prompt
3	Google Gemini Chat Model	Generates relevant and optimized hashtags

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
 - Google AI Studio API Key
 - Google Gemini Chat Model configured
 - Internet connection
-

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Hashtag Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's request.

Example Input

Generate hashtags for a LinkedIn post about Artificial Intelligence and Automation.

Output

```
{  
  "chatInput": "Generate hashtags for a LinkedIn post about Artificial Intelligence and Automation."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Hashtag Generator.

Your task is to generate relevant, engaging, and platform-appropriate hashtags based on the user's topic or content.

Instructions:

- Understand the user's topic or post.
 - Generate 15–20 relevant hashtags.
 - Include a mix of popular, niche, and trending hashtags where appropriate.
 - Avoid duplicate hashtags.
 - Ensure hashtags are suitable for the requested platform (Instagram, LinkedIn, X, YouTube, etc.).
 - Return only the hashtags.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)

Setting	Value
---------	-------

Authentication	Google AI Studio API Key
----------------	--------------------------

Connect it to the AI Agent.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Generate hashtags for an Instagram post about healthy eating.

↓

Step 2

Chat Trigger receives the request.

↓

Step 3

AI Agent understands the request.

↓

Step 4

The prompt is sent to the Google Gemini Chat Model.

↓

Step 5

Gemini generates relevant hashtags.



Step 6

The hashtags are returned to the user.

Sample Inputs and Outputs

Example 1

Input

Generate hashtags for Artificial Intelligence.

Output

#ArtificialIntelligence #AI #MachineLearning #DeepLearning #Automation #Innovation
#Technology #FutureTech #DataScience #GenerativeAI #TechTrends #AIInnovation
#DigitalTransformation #SmartTechnology #TechCommunity

Example 2

Input

Generate hashtags for Digital Marketing.

Output

#DigitalMarketing #Marketing #SEO #ContentMarketing #SocialMedia #Branding
#EmailMarketing #BusinessGrowth #MarketingStrategy #OnlineBusiness #MarketingTips
#Entrepreneur #Advertising #LeadGeneration #GrowthMarketing

Example 3

Input

Generate hashtags for Travel Photography.

Output

#Travel #TravelPhotography #Adventure #Explore #Nature #Photography #Wanderlust
#TravelGram #Vacation #Discover #PhotoOfTheDay #BeautifulDestinations #Landscape
#TravelBlogger #InstaTravel

Example 4

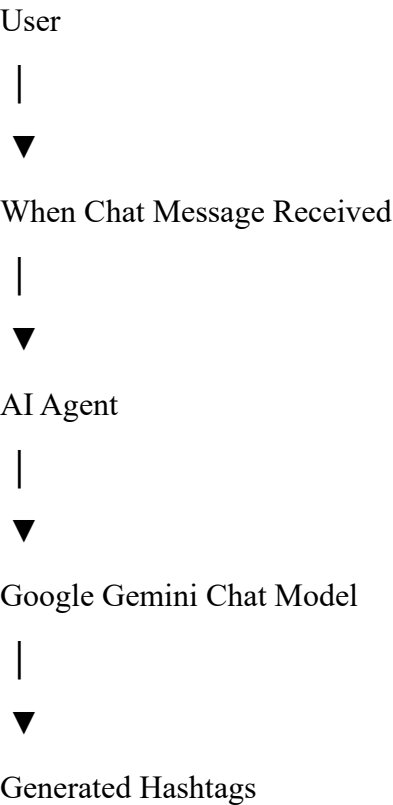
Input

Generate hashtags for Fitness Motivation.

Output

#Fitness #Workout #Gym #HealthyLifestyle #FitLife #FitnessMotivation #Health #Exercise
#Strength #Wellness #Training #BodyGoals #FitnessJourney #Motivation #StayHealthy

Workflow Diagram



Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive a user request.

Solution

Verify that the Prompt field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Hashtag Generator in n8n.
 - Configure an AI Agent with Google Gemini.
 - Design prompts for hashtag generation.
 - Generate platform-specific hashtags automatically.
 - Build conversational AI workflows for social media content.
-

Real-World Applications

- Social media marketing
- Instagram content creation
- LinkedIn post optimization
- X (Twitter) engagement
- YouTube video promotion
- Brand awareness campaigns
- Digital marketing

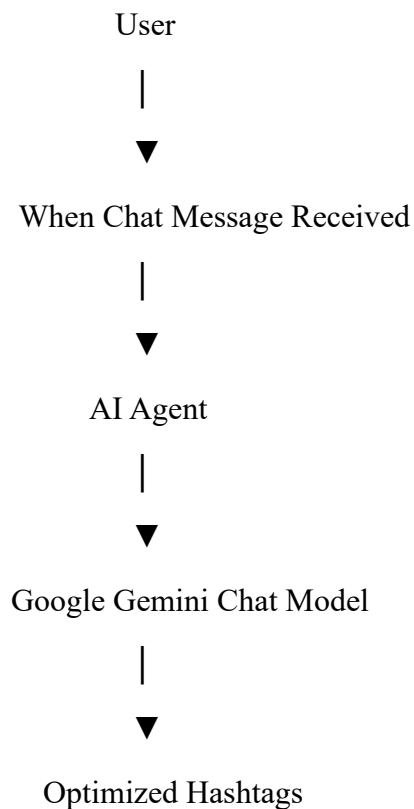
- Influencer marketing
 - Content strategy
 - Online business promotion
-

Practice Challenge

Enhance this project by adding:

1. A platform selector (Instagram, LinkedIn, X, Facebook, YouTube).
 2. A hashtag quantity selector (10, 20, 30 Hashtags).
 3. A hashtag category selector (Trending, Niche, Mixed).
 4. A Memory node to remember the user's preferred hashtag style.
 5. A Google Sheets node to save generated hashtags.
 6. A social media publishing node (or HTTP Request node) to automatically attach hashtags to scheduled posts.
-

Final Workflow



Expected Result

The **AI Hashtag Generator** accepts a topic, keyword, or social media post from the user, uses the **Google Gemini Chat Model** to generate relevant, engaging, and platform-specific hashtags, and returns a list of optimized hashtags that help improve content visibility and audience engagement. This project teaches prompt engineering, AI-powered content generation, and conversational workflow design in n8n, making it an excellent foundation for building advanced social media automation workflows.